1. Integration

1.1. Basic Integrals

ax^n	$a\frac{x^{n+1}}{(n+1)}+c$
e^{ax+b}	$\frac{1}{a}e^{ax+b}$
$\frac{1}{ax+b}$	$\frac{1}{a}\ln\lvert ax+b\rvert$
$\sin(ax+b)$	$-\frac{1}{a}\cos(ax+b)$
$\cos(ax+b)$	$\frac{1}{a}\sin(ax+b)$
$\sec^2(ax+b)$	$\frac{1}{a}\tan(ax+b)$
$(ax+b)^n$	$\frac{(ax+b)^{n+1}}{a(n+1)}$
$\frac{1}{x^2+a^2}$	$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)$

- Integration reverses a differentiation. It is the reverse of differentiation.
- Use trigonometrical relationships to facilitate complex trigonometric integrals.
- Integrate by decomposing into partial fraction.

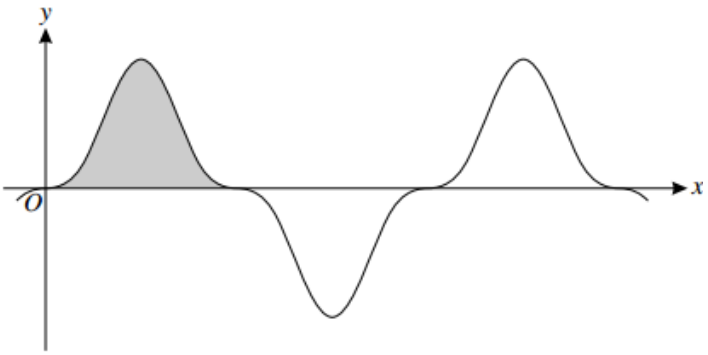
1.2. Integration by u-Substitution

$$\int f(x)\,dx=\int f(x)\frac{\mathrm{dx}}{\mathrm{du}}\,\mathrm{du}$$

- **Make x equal to something:** when differentiated, multiply the substituted form directly
- **Make u equal to something:** when differentiated, multiply the substituted form with its reciprocal
- With definite integrals, change limits in terms of u

{W12-P33} Question 7:

The diagram shows part of curve $y=\sin^3 2x\cos^3 2x$. The shaded region shown is bounded by the curve and the x-axis and its exact area is denoted by A.



Use the substitution $u=\sin 2x$ in a suitable integral to find the value of A

Solution:

To find the limit, you are trying to the find the points at which $y=0$

$\sin x=0$ at $x=0,\ \pi,\ 2\pi$ $\cos x=0$ at $x=\frac{\pi}{2},\frac{3\pi}{4}$

Choose the two closest to 0 *because the shaded area has gone through* $y=0$ *only twice*

$$\therefore 0 \text{ and } \frac{\pi}{2}$$

Since it is $\sin 2x$ *and* $\cos 2x$ *, divide both by 2*

$$\therefore \text{ Limits are } 0 \text{ and } \frac{\pi}{4}$$

Integrate by u *substitution, let:*

$$u=\sin 2x\ \frac{\mathrm{du}}{\mathrm{dx}}=2\cos 2x\ \frac{\mathrm{dx}}{\mathrm{du}}=\frac{1}{2\cos 2x}$$

$$\sin^3 2x\cos^3 2x\equiv(\sin 2x)^3\left(\cos^2 2x\right)\cos 2x$$

$$\equiv\left(\sin^3 2x\times\left(1-\sin^2 2x\right)\right)\cos 2x$$

$$\equiv\left(\sin^3 2x-\sin^5 2x\right)\cos 2x\times\frac{1}{2\cos 2x}$$

$$\equiv\frac{1}{2}\left(u^3-u^5\right)$$

Now integrate:

$$\frac{1}{2}\int\left(u^3-u^5\right)=\frac{1}{2}\left(\frac{u^4}{4}-\frac{u^6}{6}\right)$$

The limits are $x=0$ *and* $x=\frac{\pi}{4}$ *. In terms of* u*,*

$u=\sin 2(0)=0$ and $u=\sin 2\left(\frac{\pi}{4}\right)=1$

Substitute limits

$$\frac{1}{2}\left(\frac{1^4}{4}-\frac{1^6}{6}\right)-\frac{1}{2}\left(\frac{0^4}{4}-\frac{0^6}{6}\right)=\frac{1}{24}$$

1.3. Integrating $\frac{\mathbf{f}'(\mathbf{x})}{\mathbf{f}(\mathbf{x})}$

$$\int\frac{\mathbf{f}'(\mathbf{x})}{\mathbf{f}(\mathbf{x})}dx=\ln\lvert f(x)\rvert+k$$

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{S10-P32} Question 10:

By splitting into partial fractions, show that:

$$\int_1^2 \frac{2x^3-1}{x^2(2x-1)}\mathrm{dx} = \frac{3}{2} + \frac{1}{2}\ln\left(\frac{16}{27}\right)$$

Solution:

Write as partial fractions

$$\int_1^2 \frac{2x^3-1}{x^2(2x-1)}\mathrm{dx} \equiv \int_1^2 1 + \frac{2}{x} + \frac{1}{x^2} + \frac{3}{2x-1}\mathrm{dx}$$

$$\equiv x + 2\ln x - x^{-1} - \frac{3}{2}\ln|2x-1|$$

Substitute the limits

$$2 + 2\ln 2 - \frac{1}{2} - \frac{3}{2}\ln 3 - 1 - 2\ln 1 + 1 + \frac{3}{2}\ln 1$$

$$\frac{3}{2} + \frac{1}{2}\ln 16 + \frac{1}{2}\ln\frac{1}{3^3} \equiv \frac{3}{2} + \frac{1}{2}\ln\frac{16}{27}$$

1.4. Integrating By Parts

$$\int u\frac{\mathrm{dv}}{\mathrm{dx}}\,dx = uv - \int v\frac{\mathrm{du}}{\mathrm{dx}}\mathrm{dx}$$

For a definite integral:

$$\int_a^b u\frac{\mathrm{dv}}{\mathrm{dx}}\,\mathrm{dx} = [uv]_a^b - \int_a^b v\frac{\mathrm{du}}{\mathrm{dx}}\mathrm{dx}$$

What to make $\mathbf{u}$:

L	A	T	E
Logs	Algebra	Trig	*e*

{W13-P31} Question 3:

Find the exact value of

$$\int_1^4 \frac{\ln x}{\sqrt{x}}\mathrm{dx}$$

Solution:

Convert to index form:

$$\frac{\ln x}{\sqrt{x}} = x^{\frac{1}{2}}\ln x$$

Integrate by parts, let:

$$u = \ln x \quad \frac{\mathrm{du}}{\mathrm{dx}} = \frac{1}{x} \quad \frac{\mathrm{dv}}{\mathrm{dx}} = x^{-\frac{1}{2}} \qquad v = 2x^{\frac{1}{2}}$$

$$\therefore \ln x 2x^{\frac{1}{2}} - \int 2x^{\frac{1}{2}} \times x^{-1} \equiv 2\sqrt{x}\ln x - \int 2x^{-\frac{1}{2}} \equiv 2\sqrt{x}\ln x - 4\sqrt{x}$$

Substitute limits

$$= 4\ln 4 - 4$$

1.5. Integrating Powers of Sine or Cosine

To integrate $\sin x$ or $\cos x$ with a power:

- If power is odd, pull out a $\sin x$ or $\cos x$ and use Pythagorean identities and double angle identities
- If power is even, use the following identities

$$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos(2x)$$

$$\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos(2x)$$

1.6. Integrating $\cos^m x \ \sin^n x$

If $\mathbf{m}$ or $\mathbf{n}$ are odd and even, then:

- Factor out one power from **odd trig function**
- Use Pythagorean identities to transform remaining **even trig function** into the **odd trig function**
- Let u equal to **odd trig function** and integrate

If $\mathbf{m}$ and $\mathbf{n}$ are both even, then:

- Replace all even powers using the double angle identities and integrate

If $\mathbf{m}$ and $\mathbf{n}$ are both odd, then:

- Choose one of the trig. functions & factor out one power
- Use Pythagorean identity to transform remaining even power of chosen trig function into other trig. function

If either $\mathbf{m}$ or $\mathbf{n}$ or both = 1, then:

- Let u equal to the trig function whose power doesn't equal 1 then integrate
- If both are 1, then let u equal either

{W09-P31} Question 5:

i) Prove the identity

$$\cos 4\theta - 4\cos 2\theta + 3 \equiv 8\sin^4\theta$$

ii) Using this result find, in simplified form, the exact value of

$$\int_{\frac{1}{6}\pi}^{\frac{1}{3}\pi} \sin^4\theta\mathrm{d}\theta$$

Solution:

Part (i)

Use double angle identities

$$\cos 4\theta - 4\cos 2\theta + 3$$

$$\equiv 1 - 2\sin^2 2\theta - 4\left(1 - 2\sin^2\theta\right) + 3$$

Open everything and clean

$$\equiv 1 - 2\sin^2 2\theta - 4 + 8\sin^2\theta + 3$$

$$\equiv 1 - 2(\sin 2\theta)^2 - 4 + 8\sin^2\theta + 3$$

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$$\equiv 1 - 2(2\sin\theta\cos\theta)^2 - 4 + 8\sin^2\theta + 3$$

$$\equiv 1 - 2\left(4\sin^2\theta\cos^2\theta\right) - 4 + 8\sin^2\theta + 3$$

$$\equiv 1 - 2(4\sin^2\theta(1-\sin^2\theta)) - 4 + 8\sin^2\theta + 3$$

$$\equiv 1 - 8\sin^2\theta + 8\sin^4\theta - 4 + 8\sin^2\theta + 3$$

$$\equiv 8\sin^4\theta$$

Part (ii)

Use identity from (part i):

$$\frac{1}{8}\int_{\frac{1}{6}\pi}^{\frac{1}{3}\pi} \cos 4\theta - 4\cos 2\theta + 3$$

$$\equiv \frac{1}{8}\left[\frac{1}{4}\sin 4\theta - 2\sin\theta + 3\theta\right]_{\frac{1}{6}\pi}^{\frac{1}{3}\pi}$$

Substitute limits

$$\equiv \frac{1}{32}\left(2\pi - \sqrt{3}\right)$$

{W12-P32} Question 5:

i. By differentiating $\frac{1}{\cos x}$, show that if $y = \sec x$ then $\frac{\mathrm{dy}}{\mathrm{dx}} = \sec x\tan x$

ii. Show that $\frac{1}{\sec x - \tan x} \equiv \sec x + \tan x$

iii. Deduce that:

$$\frac{1}{(secx - \tan x)^2} \equiv 2\sec^2 x - 1 + 2\sec x\tan x$$

iv. Hence show that:

$$\int_0^{\frac{1}{4}\pi} \frac{1}{(secx - \tan x)^2} dx = \frac{1}{4}\left(8\sqrt{2} - \pi\right)$$

Solution:

Part (i)

Change to index form:

$$\frac{1}{\cos x} = \cos^{-1} x$$

Differentiate by chain rule:

$$\frac{\mathrm{dy}}{\mathrm{dx}} = -1(\cos x)^{-2} \times (-\sin x)$$

$$-1(\cos x)^{-2} \times (-\sin x) \equiv \frac{\sin x}{\cos^2 x} \equiv \frac{\sin x}{\cos x} \times \frac{1}{\cos x}$$

$$\frac{\sin x}{\cos x} \times \frac{1}{\cos x} \equiv \sec x\tan x$$

Part (ii)

Multiply numerator and denominator by $\sec x + \tan x$

$$\frac{\sec x + \tan x}{(secx - \tan x)(\sec x + \tan x)} \equiv \frac{\sec x + \tan x}{\sec^2 x - \tan^2 x}$$

$$\frac{\sec x + \tan x}{\sec^2 x - \tan^2 x} \equiv \frac{\sec x + \tan x}{1} \equiv \sec x + \tan x$$

Part (iii)

Substitute identity from (part ii)

$$\frac{1}{(secx - \tan x)^2} \equiv (secx + \tan x)^2$$

Open out brackets

$$(secx + \tan x)^2$$

$$\equiv \sec^2 x + 2\sec x\tan x + \tan^2 x$$

$$\equiv \sec^2 x + 2\sec x\tan x + \sec^2 x - 1$$

$$\equiv 2sec^2 x + 2\sec x\tan x - 1$$

$$\equiv 2\sec^2 x - 1 + 2\sec x\tan x$$

Part (iv)

$$\int \frac{1}{(secx - \tan x)^2}\mathrm{dx}$$

$$\equiv \int 2\sec^2 x - 1 + 2\sec x\tan x\mathrm{dx}$$

$$\equiv 2\int \sec^2 x - \int 1 + 2\int \sec^2 x\tan^2 x$$

Using differential from part i:

$$\equiv 2\tan x - x + 2\sec x$$

Substitute boundaries:

$$= \frac{1}{4}\left(8\sqrt{2} - \pi\right)$$

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